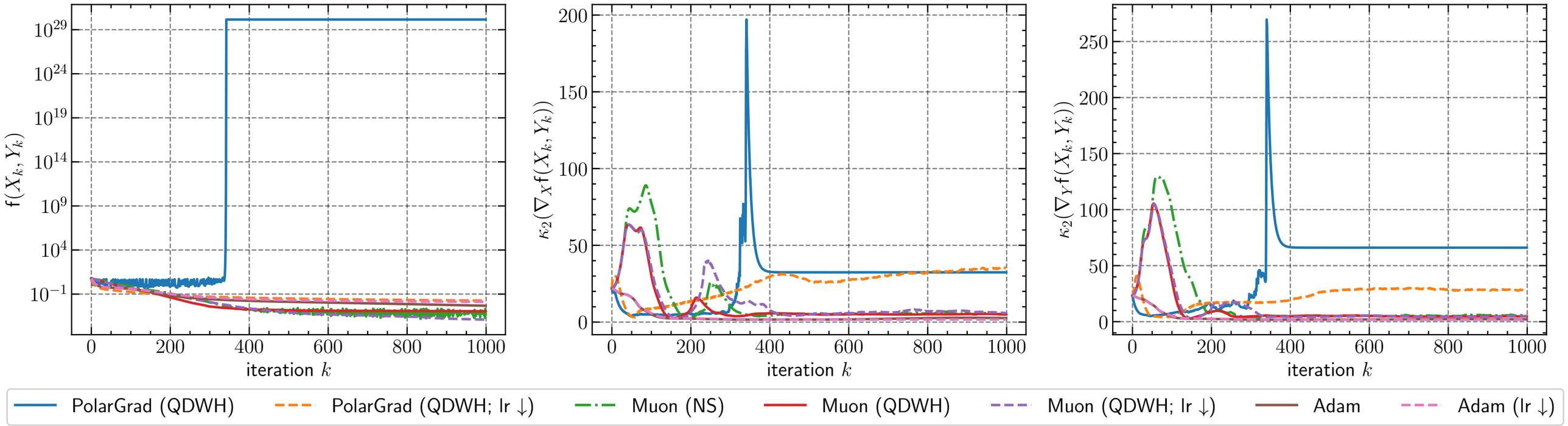
10^{29}
10^{24}
10^{19}
10^{14}
10^{9}
10^{4}
10^{-1}
$\mathsf{f}(X_k, Y_k)$
0
200
400
600
800
1000
iteration k
200
150
100
50
0
$\kappa_2(\nabla_X \mathsf{f}(X_k, Y_k))$
0
200
400
600
800
1000
iteration k
250
200
150
100
50
0
$\kappa_2(\nabla_Y \mathsf{f}(X_k, Y_k))$
0
200
400
600
800
1000
iteration k
PolarGrad (QDWH)
PolarGrad (QDWH; lr ↓)
Muon (NS)
Muon (QDWH)
Muon (QDWH; lr ↓)
Adam
Adam (lr ↓)